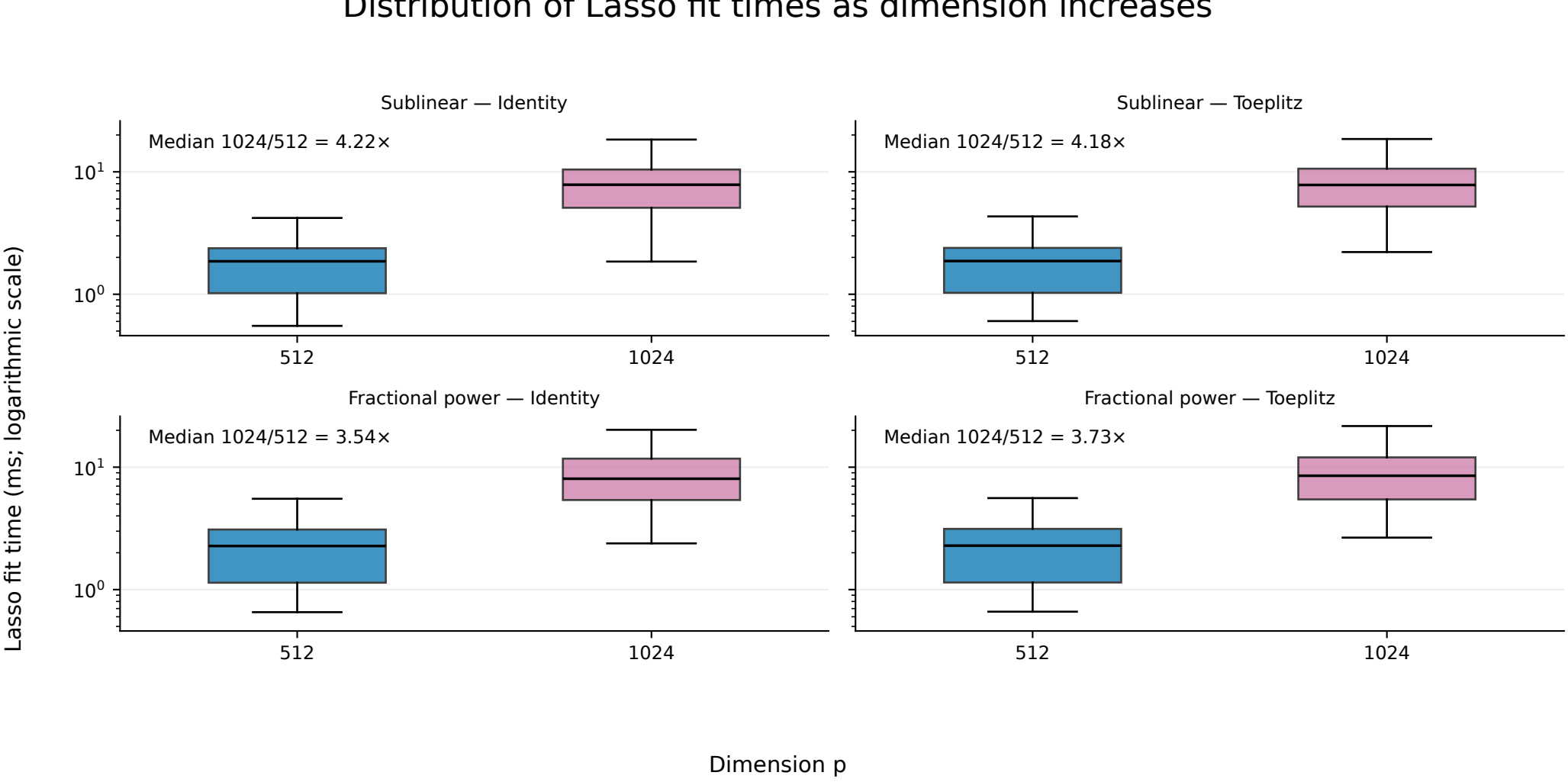

Each box pools 2,400 fits over θ = 0.1, 0.3, …, 2.3 (200 trials per value); both n and k grow with p.
Gaussian predictors: independent (left) or Toeplitz, adjacent correlation 0.1 (right); noise SD = 0.5; active magnitudes = 1.
$n = \text{floor}[2\theta k \log(p - k)] + k + 1$; $\alpha = 0.4$, $\gamma = 0.75$. Timing includes fitting only, on the same machine and software.
Boxes: middle 50%; line: median; whiskers: data within 1.5 × IQR fences. Outlier markers hidden; all fits retained in summaries.
Boxes describe variability among fits, not confidence intervals for the medians.